



Addis Ababa Science and Technology University
Department of Electromechanical Engineering

Internship Report

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Submitted to:

Department of Electromechanical Engineering, AASTU

Duration:

[Start Date] to [End Date]

Organization:

[Company Name]

[Company Location]

August 1, 2025

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1 Introduction and Company Profile

1.1 Introduction

Instruction: Briefly state the purpose of your internship (200-300 words). Explain how it relates to your electromechanical engineering studies and what you hoped to achieve.

Example (Individual): My internship at EthioRobotics focused on maintenance of automated packaging lines, allowing me to apply knowledge from courses like MEng 3201 (Control Systems) and MEng 3105 (Electromechanical Devices). I aimed to develop practical skills in PLC troubleshooting and mechanical system maintenance.

Example (Group): Our team's internship at Addis Automation involved designing a conveyor control system. This project integrated our knowledge from MEng 4203 (Industrial Automation) and MEng 4102 (Power Electronics). We sought to gain experience in collaborative engineering projects and system integration.

1.2 Company Profile

Instruction: Provide a concise overview (1-2 pages) covering:

- Company's core activities relevant to electromechanical systems
- Size, location, and market position
- Key electromechanical products/processes
- Organizational structure

Example: TechnoElectro PLC (Addis Ababa) specializes in industrial motor control systems with 120 employees. Their main product line includes variable frequency drives (VFDs) for Ethiopian textile factories. The company has three technical departments: Mechanical Design, Electrical Systems, and Control Engineering, working together on integrated solutions.

2 Internship Activities and Responsibilities

2.1 Department Overview

Instruction: Describe your assigned department (150-200 words), focusing on its role in electromechanical systems.

Example: The Maintenance Engineering Department where I was assigned handles all electromechanical systems in the production facility. The team consists of 5 engineers and 12 technicians responsible for preventive maintenance, troubleshooting, and upgrades of conveyor systems, robotic arms, and packaging machines.

2.2 Daily Responsibilities

Instruction: List your routine tasks (bullet points):

- Focus on electromechanical aspects

- Include both design and maintenance activities
- Specify tools/equipment used

Example:

- Calibrated servo motors using Allen-Bradley PowerFlex drives
- Troubleshoot PLC (Siemens S7-1200) ladder logic for faulty sensors
- Performed vibration analysis on conveyor rollers using Fluke 810
- Documented maintenance procedures for pneumatic systems

2.3 Key Projects

Instruction: For each major project (1-2 paragraphs per project):

- Problem statement and electromechanical components involved
- Your specific technical contributions
- System diagrams/schematics (include in appendices)

Example: Conveyor System Upgrade

- **Problem:** Outdated conveyor causing 15% product damage
- **Solution:** Upgraded to servo-driven system with:
 - Yaskawa servo motors (mechanical)
 - Omron proximity sensors (electrical)
 - Custom PID control algorithm (software)
- **My Role:** Designed motor mounting brackets and assisted in PID tuning

3 Technical Skills and Tools Used

3.1 Hardware Tools

Instruction: List electromechanical equipment used:

- Industrial controllers (PLCs, microcontrollers)
- Sensors and actuators
- Power systems and drives
- Measurement tools

Example:

- Siemens S7-1200 PLC
- Yaskawa servo drives
- SICK photoelectric sensors
- Fluke 87V multimeter

3.2 Software Tools

Instruction: Mention:

- CAD/CAM software (e.g., SolidWorks, AutoCAD Electrical)
- Programming environments (e.g., Ladder Logic, MATLAB)
- Simulation tools

Example:

- Siemens TIA Portal (PLC programming)
- SolidWorks (mechanical design)
- MATLAB/Simulink (control system simulation)

3.3 Technical Skills Gained

Instruction: Highlight 3-5 key electromechanical competencies developed.

Example:

- PLC troubleshooting and programming
- Mechanical system vibration analysis
- Electromechanical system integration
- Industrial safety protocols

4 Challenges and Key Learnings

4.1 Challenges Faced

Instruction: Describe 2-3 technical challenges (e.g., system integration issues, troubleshooting electromechanical systems).

Example:

- Signal interference causing false sensor readings
- Mechanical misalignment in servo-driven systems
- Integrating legacy systems with new controllers

4.2 Problem-Solving Approaches

Instruction: Explain how you applied electromechanical engineering principles to resolve challenges.

Example: For signal interference:

1. Identified noise sources using spectrum analyzer
2. Implemented shielded cabling
3. Added ferrite cores to sensitive circuits
4. Redesigned grounding system

4.3 Key Learnings

Instruction: Reflect on:

- Industry vs. academic knowledge differences
- Safety practices in electromechanical systems
- Interdisciplinary teamwork experiences

Example:

- Practical systems require more robust error handling than theoretical models
- Documentation is crucial for maintenance and troubleshooting
- Effective communication between mechanical and electrical teams is essential

5 Summary and Recommendations

5.1 Summary

Instruction: Concisely summarize your overall experience (150 words).

Example: This internship provided invaluable practical experience in industrial electromechanical systems. I gained hands-on skills in PLC programming, mechanical system maintenance, and system integration that complemented my academic studies. The experience highlighted the importance of interdisciplinary collaboration and practical problem-solving in real-world engineering applications.

5.2 Recommendations

Instruction: Provide:

- Suggestions for the company's internship program
- Curriculum improvement ideas for AASTU's electromechanical program
- Career advice for future interns

Example:

- **For Company:** Implement structured mentorship pairings
- **For AASTU:** Add more industrial communication protocols to curriculum
- **For Future Interns:** Take initiative to learn from technicians' practical knowledge

6 Appendices

Instruction: Include:

- Technical drawings/diagrams (properly labeled)
- Data sheets (if permitted)
- Photos of relevant electromechanical systems
- Weekly activity logs

Example Appendix Items:

- Appendix A: Conveyor system schematic
- Appendix B: PLC program listing
- Appendix C: Vibration analysis reports

7 References

Instruction: Use IEEE format for all citations. Include:

- Technical manuals referenced
- Company documentation (if allowed)
- Academic references

Example References:

- Siemens S7-1200 Programmable Controller Manual (2022)
- Industrial Automation Handbook (Tekle, 2021)
- Safety Protocols (2023)